

ECx00M&ECx00N&EC800K Series

QuecCell Application Note

LTE Standard Module Series

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Quectel Wireless Solutions Co., Ltd.

Building 5, Shanghai Business Park Phase III (Area B), No.1016 Tianlin Road, Minhang District, Shanghai 200233, China

Tel: +86 21 5108 6236

Email: info@quectel.com

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About the Document

Revision History

Version	Date	Author	Description
-	2023-04-10	Sawyer HE	Creation of the document
1.0	2023-04-10	Sawyer HE	First official release
1.1	2023-12-20	Sawyer HE	Added applicable module EC800K series.

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1 Introduction

This document is an application note for AT commands related to QuecCell on Quectel ECx00M family, ECx00N family and EC800K series modules.

QuecCell is a featured function embedded in Quectel modules. It can be used to scan the detailed information of base stations.

1.1. Applicable Modules

Table 1: Applicable Modules

Module Family	Module
ECx00M	EC200M-CN
	EC600M-CN
	EC800M-CN
ECx00N	EC200N-CN
	EC800N-CN
-	EC800K Series

2 Description of AT Commands

2.1. AT Command Introduction

2.1.1. Definitions

- **<CR>** Carriage return character.
- **<LF>** Line feed character.
- **<...>** Parameter name. Angle brackets do not appear on command line.
- **[...]** Optional parameter of a command or an optional part of TA information response. Square brackets do not appear on the command line. When an optional parameter is not given in a command, the new value equals its previous value or the default settings, unless otherwise specified.
- **Underline** Default setting of a parameter.

2.1.2. AT Command Syntax

All command lines must start with **AT** or **at** and end with **<CR>**. Information responses and result codes always start and end with a carriage return character and a line feed character: **<CR><LF><response><CR><LF>**. In tables presenting commands and responses throughout this document, only the commands and responses are presented, and **<CR>** and **<LF>** are deliberately omitted.

Table 2: Types of AT Commands

Command Type	Syntax	Description
Test Command	AT+<cmd>=?	Test the existence of the corresponding command and return information about the type, value, or range of its parameter.
Read Command	AT+<cmd>?	Check the current parameter value of the corresponding command.
Write Command	AT+<cmd>=<p1>[,<p2>[,<p3>[...]]]	Set user-definable parameter value.
Execution Command	AT+<cmd>	Return a specific information parameter or perform a specific action.

2.2. Declaration of AT Command Examples

The AT command examples in this document are provided to help you familiarize with AT commands and learn how to use them. The examples, however, should not be taken as Quectel's recommendation or suggestions about how you should design a program flow or what status you should set the module into. Sometimes multiple examples may be provided for one AT command. However, this does not mean that there exists a correlation among these examples and that they should be executed in a given sequence.

2.3. AT+QENG Get Information of Primary Serving Cell and Neighbour Cells

This command gets the information of the primary serving cell and neighbour cells.

AT+QENG Get Information of Primary Serving Cell and Neighbour Cells	
Test Command AT+QENG=?	Response +QENG: (list of supported <cell_type>s) OK
Write Command AT+QENG="servingcell"	Response In the case of LTE mode: +QENG: "servingcell",<state>,"LTE",<is_tdd>,<MCC>,<MNC>,<cellID>,<PCI>,<EARFCN>,<freq_band_ind>,<UL_bandwidth>,<DL_bandwidth>,<TAC>,<RSRP>,<RSRQ>,<RSSI>,<SINR>,<srxlev> OK In the case of WCDMA mode: +QENG: "servingcell",<state>,"WCDMA",<MCC>,<MNC>,<LAC>,<cellID>,<UARFCN>,<PSC>,<RAC>,<RSCP>,<ecno>,-,-,-,- OK In the case of GSM mode: +QENG: "servingcell",<state>,"GSM",<MCC>,<MNC>,<LAC>,<cellID>,<BSIC>,<ARFCN>,<GSM_band>,<rxlev>,<TX_power>,<rla>,<DRX>,<c1>,<c2>,<GPRS_support>,-,-,-,-,-,-,-,-,-,- OK

	OK
Write Command AT+QENG="neighbourcell"	Response In the case of LTE mode: [+QENG: "neighbourcell intra","LTE",<EARFCN>,<PCI>,<RSRP>,<RSRQ>,-,<srlev>,<cell_resel_priority>,<s_no n_intra_search>,<thresh_serving_low>,<s_intra_search> [...]] [+QENG: "neighbourcell inter","LTE",<EARFCN>,<PCI>,<RSRP>,<RSRQ>,-,<srlev>,<threshX_low>,<threshX_high>,<cell_resel_priority> [...]] [+QENG: "neighbourcell","GSM",<ARFCN>,<cell_resel_priority>,<thresh_GSM_high>,<thresh_GSM_low>,<NCC_permitted>,<GSM_band>,<BSIC>,<RSSI>,<rxlev> [...]] [+QENG: "neighbourcell","WCDMA",<UARFCN>,<cell_resel_priority>,<threshX_high>,<threshX_low>,<PSC>,<CPICH_rscp>,<CPICH_ecno>,<srlev> [...]] OK In the case of WCDMA mode: [+QENG: "neighbourcell","WCDMA",<UARFCN>,<srqual>,<PSC>,<RSCP>,<ecno>,<set>,-,<srlev> [...]] [+QENG: "neighbourcell","GSM",<BSIC>,<RSSI>,<rxlev>,<rank> [...]] [+QENG: "neighbourcell","LTE",<EARFCN>,<PCI>,<RSRP>,<RSRQ>,<srlev> [...]] OK In the case of GSM mode: [+QENG: "neighbourcell","GSM",<MCC>,<MNC>,<LAC>,<cellID>,<BSIC>,<ARFCN>,<rxlev>,<c1>,<c2>,<c31>,<c32> [...]] [+QENG: "neighbourcell","WCDMA",<UARFCN>,<PSC>,<RSCP>,<ecno> [...]] [+QENG: "neighbourcell","LTE",<EARFCN>,<PCI>,<RSR

	P>,<RSRQ> [...]
	OK
Write Command AT+QENG="3gcomm"	Response Only in WCDMA network. It gets the common information on 3G cell, including the information on 3G serving cell, 3G neighbour cell and 2G neighbour cell. WCDMA serving cell information: [+QENG: "3gcomm",<cell_type>,<RAT>,<state>,<MCC>,<MNC>,<LAC>,<cellID>,<UARFCN>,<PSC>,<RSSI>,<RSCP>,<ecno>,-,- [...] 3G neighbour cell information of WCDMA serving cell: [+QENG: "3gcomm",<cell_type>,<RAT>,-,-,-,<UARFCN>,<PSC>,<RSSI>,<RSCP>,<ecno>,<srxqual>,<srxlev> [...] 2G neighbour cell information of WCDMA serving cell: [+QENG: "3gcomm",<cell_type>,<RAT>,<ARFCN>,<BSIC>,<RSSI>,<rxlev>,<rank> [...] OK
Maximum Response Time	300 ms
Characteristics	/

Parameter

<cell_type>	String type. Information of different cells. "servingcell" Information of 2G/3G/4G serving cells "neighbourcell" Information of 2G/3G/4G neighbour cells "3gcomm" Information of 3G serving cell/2G neighbour cell/3G neighbour cell in WCDMA network
<state>	String type. UE state. "SEARCH" UE is searching but could not (yet) find a suitable 2G/3G/4G cell "LIMSRV" UE is camping on a cell but has not registered on the network "NOCONN" UE has camped on a cell and registered on the network, and it is in idle state "CONNECT" UE has camped on a cell and registered on the network, and

	a call is in progress
<RAT>	String type. Radio access technologies. It includes "2G", "3G" and "4G".
<is_tdd>	String type. Communication mode. "TDD" Time division duplex mode "FDD" Frequency division duplex mode
<MCC>	Integer type. Mobile country code (first part of the PLMN code).
<MNC>	Integer type. Mobile network code (second part of the PLMN code).
<cellID>	Hexadecimal format. Cell ID. 16-bit (GSM), 28-bit (UMTS), or 32-bit (LTE) cell ID. Range: 0–0xFFFFFFFF.
<PCI>	Integer type. Physical cell identity. Range: 0–503.
<EARFCN>	Integer type. E-UTRA absolute radio frequency channel number. Range: 0–65535. (If new bands are supported, the range of the parameter for these bands is 65536–70645.)
<freq_band_ind>	Integer type. E-UTRA frequency band. (See 3GPP 36.101.)
<UL_bandwidth>	Integer type. Uplink bandwidth. 0 1.4 MHz 1 3 MHz 2 5 MHz 3 10 MHz 4 15 MHz 5 20 MHz
<DL_bandwidth>	Integer type. Downlink bandwidth. 0 1.4 MHz 1 3 MHz 2 5 MHz 3 10 MHz 4 15 MHz 5 20 MHz
<TAC>	Hexadecimal format. Tracking area code. Range: 0–0xFFFFFFFF. (See 3GPP 23.003 subclause 19.4.2.3.)
<RSRP>	Integer type. Reference signal received power. Unit: dBm. (See 3GPP 36.214 subclause 5.1.1.)
<RSRQ>	Integer type. Reference signal received quality. Unit: dB. (See 3GPP 36.214 subclause 5.1.3.)
<RSSI>	Integer type. Received signal strength indication. Unit: dBm.
<SINR>	Integer type. Signal to interference plus noise ratio. Range: -20 to 30. Unit: dB.
<srxlev>	Integer type. Cell selection Rx level value (in dB). (See 3GPP 25.304/3GPP 36.304.)
<LAC>	Hexadecimal format. Location area code. The two-byte location area code (e.g., 00C1 equals 193 in decimal) of the cell that is scanned. Range: 0–0xFFFF.
<UARFCN>	Integer type. UTRA radio frequency channel number. Range: 712–10838.
<PSC>	Integer type. Primary scrambling code. Range: 0–511.

<RAC>	Integer type. Routing area code. Range: 0–255.
<RSCP>	Integer type. Received signal code power.
<ecno>	Integer type. EC/NO is the ratio of the received energy per PN chip to the total received power spectral density, which is relevant to the channel. Unit: dB.
<BSIC>	Integer type. Base station identity code. Range: 0–63.
<ARFCN>	Integer type. Absolute radio frequency channel number. Range: 0–1023.
<GSM_band>	Integer type. GSM frequency band. 0 DCS1800 1 PCS1900 2 GSM900 "- " Other GSM frequency bands
<rxlev>	Integer type. RX level in the cell. Range: 0–63. Unit: dB. Subtracting 111 from the RX level value, a dBm value will be got. (See 3GPP 25.304.)
<TX_power>	Integer type. TX power level.
<rla>	Integer type. Minimum access RX level.
<DRX>	Integer type. Discontinuous reception cycle.
<c1>	Integer type. Cell selection criterion.
<c2>	Integer type. Cell reselection criterion.
<GPRS_support>	Integer type. It indicates whether the current cell supports GPRS or not. 0 Not support 1 Support
<cell_resel_priority>	Integer type. Cell reselection priority. Range: 0–7.
<s_non_intra_search>	Integer type. Threshold to control non-intra frequency search.
<thresh_serving_low>	Integer type. The <srxlev> threshold (in dB) used by the UE on the serving cell when reselecting towards a lower priority RAT/frequency.
<s_intra_search>	Integer type. Measurement trigger threshold for same-frequency cell reselection.
<threshX_low>	Integer type. It can be used as a reference when reselecting towards a lower priority RAT/frequency than the current serving frequency. The suitable receive level value of an evaluated lower priority cell must be greater than this value. Range: 0–31. Unit: 2 dB.
<threshX_high>	Integer type. It can be used as a reference when reselecting towards a higher priority RAT/frequency than the current serving frequency. The suitable receive level value of an evaluated higher priority cell must be greater than this value. Range: 0–31. Unit: 2 dB.
<thresh_GSM_high>	Integer type. The reselection threshold of higher priority cell.
<thresh_GSM_low>	Integer type. The reselection threshold of lower priority cell.
<NCC_permitted>	Integer type. It specifies whether to report the bitmask of neighbour cell with specific network color code.
<CPICH_rscp>	Integer type. CPICH/PCCPCH received signal code power. RSCP is transmitted as integer in the range of -120 dBm to -25 dBm in UMTS FDD/TDD message. According to 3GPP 25.133, this value is encoded as integer in the range of -5 to 99.

<CPICH_ecno>	Integer type. Ratio of the received energy per PN chip for the CPICH to the total received power spectral density at UE antenna connector.
<srxqual>	Integer type. Cell selection quality value. Only for FDD. Unit: dB.
<set>	Integer type. Settings of 3G neighbour cell. 0/2 Sync Neighbour Set 1 Active Neighbour Set 3 Async Neighbour Set
<rank>	Integer type. Rank of the cell as neighbour for inter-RAT cell reselection.
<c31>	Integer type. GPRS signal level threshold criterion.
<c32>	Integer type. Cell ranking criterion which is used for selecting cells among those with the same priority.

NOTE

1. It indicates that the parameter is invalid under current conditions when "-" or - is returned.
2. 2G neighbor cells are only displayed when the module is in the idle state.
3. When the return value of <PCI> is 65535, it indicates that the parameter is invalid.
4. When the return value of <BSIC> is 255, it indicates that the parameter is invalid.
5. When the module failed to parse the SIB information of neighbour cells (including some parameters of AT+QENG="neighbourcell"), the value of the unsuccessfully parsed parameter is invalid.

Example

```

AT+QENG="servingcell"
+QENG: "servingcell","SEARCH"

OK
AT+QENG="servingcell"
+QENG: "servingcell","NOCONN","LTE","FDD",460,01,5A29C0B,465,1650,3,5,5,DE10,-69,-5,-65,21,
51

OK
AT+QENG="servingcell"
+QENG: "servingcell","NOCONN","WCDMA",460,01,D5D5,8054BBF,10633,457,10,-72,-5,-,-,-,-,-

OK
AT+QENG="servingcell"
+QENG: "servingcell","NOCONN","GSM",460,00,550B,D89,35,59,2,-82,0,8,0,21,131,1,-,-,-,-,-,-,-,
-,"-"

OK
AT+QENG="neighbourcell"
+QENG: "neighbourcell inter","LTE",38950,65535,-140,-20,-,-,0,24,24,6
+QENG: "neighbourcell inter","LTE",36275,65535,-140,-20,-,-,0,24,24,5
    
```

```
+QENG: "neighbourcell inter","LTE",37900,65535,-140,-20,-,0,24,24,5
+QENG: "neighbourcell inter","LTE",38098,65535,-140,-20,-,0,24,24,5
+QENG: "neighbourcell inter","LTE",38400,65535,-140,-20,-,0,24,24,3
+QENG: "neighbourcell","GSM",616,0,0,0,0,0,255,0,0
+QENG: "neighbourcell","GSM",612,0,0,0,0,0,255,0,0
+QENG: "neighbourcell","GSM",64,0,0,0,0,0,255,0,0
+QENG: "neighbourcell","GSM",63,0,0,0,0,0,255,0,0
```

OK

AT+QENG="neighbourcell"

```
+QENG: "neighbourcell","WCDMA",10633,-18,459,-32768,-32768,0,-,-115
+QENG: "neighbourcell","WCDMA",10633,-18,462,-32768,-32768,0,-,-115
+QENG: "neighbourcell","WCDMA",10633,-18,319,-32768,-32768,0,-,-115
+QENG: "neighbourcell","WCDMA",10633,-18,307,-32768,-32768,0,-,-115
+QENG: "neighbourcell","WCDMA",10633,-18,146,-32768,-32768,0,-,-115
+QENG: "neighbourcell","WCDMA",10633,-18,145,-32768,-32768,0,-,-115
+QENG: "neighbourcell","WCDMA",10633,-18,334,-32768,-32768,0,-,-115
```

OK

AT+QENG="3gcomm"

```
+QENG: "3gcomm","servingcell","3G","NOCONN",460,01,D5D5,8054BBF,10633,457,-61,-71,
-3,-,
+QENG: "3gcomm","neighbourcell","3G",-,-,-,10633,145,-85,-85,-17,-17,-115
```

OK

2.4. AT+QCELL Get Information of Serving Cell and Neighbour Cells

This command gets the information of the serving cell and neighbour cells.

AT+QCELL Get Information of Serving Cell and Neighbour Cells

Test Command AT+QCELL=?	Response OK
Read Command AT+QCELL?	Response In the case of LTE mode: +QCELL: "servingcell","LTE",<MCC>,<MNC>,<TAC>,<cellID>,<PCI>,<RX_lev> [+QCELL: "neighbourcell intra","LTE",<MCC>,<MNC>,<TAC>,<cellID>,<PCI>,<RX_lev> [...] [+QCELL: "neighbourcell inter","LTE",<MCC>,<MNC>,<TAC>,<cellID>,<PCI>,<RX_lev>

	[...]] [+QCELL: "neighbourcell","WCDMA",<MCC>,<MNC>,<LAC>,<cellID>,<PSC>,<RX_lev> [...]] [+QCELL: "neighbourcell","GSM",<MCC>,<MNC>,<LAC>,<cellID>,<BSIC>,<RX_lev> [...]] In the case of WCDMA mode: +QCELL: "servingcell","WCDMA",<MCC>,<MNC>,<LAC>,<cellID>,<PSC>,<RX_lev> [+QCELL: "neighbourcell","WCDMA",<MCC>,<MNC>,<LAC>,<cellID>,<PSC>,<RX_lev> [...]] [+QCELL: "neighbourcell intra","LTE",<MCC>,<MNC>,<TAC>,<cellID>,<PCI>,<RX_lev> [...]] [+QCELL: "neighbourcell inter","LTE",<MCC>,<MNC>,<TAC>,<cellID>,<PCI>,<RX_lev> [...]] [+QCELL: "neighbourcell","GSM",<MCC>,<MNC>,<LAC>,<cellID>,<BSIC>,<RX_lev> [...]] In the case of GSM mode: +QCELL: "servingcell","GSM",<MCC>,<MNC>,<LAC>,<cellID>,<BSIC>,<RX_lev> [+QCELL: "neighbourcell","GSM",<MCC>,<MNC>,<LAC>,<cellID>,<BSIC>,<RX_lev> [...]] [+QCELL: "neighbourcell","WCDMA",<MCC>,<MNC>,<LAC>,<cellID>,<PSC>,<RX_lev> [...]] [+QCELL: "neighbourcell intra","LTE",<MCC>,<MNC>,<TAC>,<cellID>,<PCI>,<RX_lev> [...]] [+QCELL: "neighbourcell inter","LTE",<MCC>,<MNC>,<TAC>,<cellID>,<PCI>,<RX_lev> [...]] OK
Maximum Response Time	300 ms
Characteristics	/

Parameter

<MCC>	Integer type. Mobile country code (the first part of the PLMN code).
<MNC>	Integer type. Mobile network code (the second part of the PLMN code).
<LAC>	Hexadecimal format. Location area code. The two-byte location area code (e.g., 00C1 equals 193 in decimal) of the cell that is scanned. Range: 0–0xffff.
<TAC>	Hexadecimal format. Tracking area code. Range: 0–0xffffffff. (See 3GPP 23.003 Section 19.4.2.3.)
<cellID>	Hexadecimal format. Cell ID. 16-bit (GSM), 28-bit (UMTS), or 32-bit (LTE) cell ID. Range: 0–0xffffffff.
<BSIC>	Integer type. Base station identity code. Range: 0–63.
<PSC>	Integer type. Primary scrambling code. Range: 0–511.
<PCI>	Integer type. Physical cell identity. Range: 0–503.
<RX_lev>	Integer type. RX level in the cell. Range: 0–63. Unit: dB. (See 3GPP 25.304.)

NOTE

This command can only be executed when the (U)SIM card is not inserted or when the module is in the idle state with the (U)SIM card inserted; it is recommended to use the command when the (U)SIM card is not inserted.

Example

```
AT+QCELL? //Get the information of serving cell and neighbour cells.
+QCELL: "servingcell","LTE",460,00,550b,d6b5c0,123,36
+QCELL: "neighbourcell inter","LTE",460,00,550b,5e05e2a,20,12
+QCELL: "neighbourcell inter","LTE",460,00,550b,5c4ef29,121,25
+QCELL: "neighbourcell","GSM",460,00,550b,d89,35,46
+QCELL: "neighbourcell","GSM",460,00,550b,34b8,10,45
OK
```

3 Appendix References

Table 3: Terms and Abbreviations

Abbreviation	Description
3GPP	3rd Generation Partnership Project
ARFCN	Absolute Radio-Frequency Channel Number
BSIC	Base Station Identity Code
CPICH	Common Pilot Channel
DL	Downlink
DRX	Discontinuous Reception
EARFCN	E-UTRA Absolute Radio Frequency Channel Number
E-UTRA	Evolved Universal Terrestrial Radio Access
FDD	Frequency Division Duplex
GPRS	General Packet Radio Service
GSM	Global System for Mobile Communications
LAC	Location Area Code
LTE	Long Term Evolution
MCC	Mobile Country Code
MNC	Mobile Network Code
NCC	Network Color Code
PCCPCH	Physical Common Control Physical Channel
PCI	Physical Cell Identity
PLMN	Public Land Mobile Network

PSC	Primary Scrambling Code
RAC	Routing Area Code
RAT	Radio Access Technology
RSCP	Received Signal Code Power
RSRP	Reference Signal Received Power
RSRQ	Reference Signal Received Quality
RSSI	Received Signal Strength Indicator
RX	Receive
SIB	System Information Block
SINR	Signal to Interference plus Noise Ratio
TA	Terminal Adapter
TAC	Tracking Area Code
TDD	Time Division Duplex
TX	Transmit
UARFCN	UTRA Absolute RF Channel Number
UE	User Equipment
UL	Uplink
UMTS	Universal Mobile Telecommunications System
UTRA	UMTS Terrestrial Radio Access
WCDMA	Wideband Code Division Multiple Access